Lakshya JEE 2.0 2025

Mathematics

Determinants

Practice Sheet

Q1 Let m be a positive integer &

$$D_r = \begin{vmatrix} 2r-1 & {}^mC_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix}$$
$$(0 \leq r \leq m),$$

then the value of $\sum_{r=0}^{m} D_r$ is given by:

(A) 0

(B) m^2-1

(C) 2^m

(D) $2^m \sin^2(2^m)$

Q2 There are two values of x making the value of the determinant $\begin{vmatrix} 1 & -2 & 5 \\ 2 & x & -1 \\ 0 & 4 & 2x \end{vmatrix}$ equal to 86 .

The sum of these two numbers, is

(A) -4 (B) 5

(C) -3 (D) 9

Q3 Let $f(x)$

$$= \begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 1+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 1+4\sin 2x \end{vmatrix}$$

then

the maximum value of $f(x) =$

(A) 2 (B) 4

(C) 6 (D) 8

Q4 If $px^4+qx^3+rx^2+sx+t$

$$= \begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & 2-x & x-3 \\ x-3 & x+4 & 3x \end{vmatrix}$$

then t $=$

(A) 33 (B) 0

(C) 21 (D) none

Q5 If $\alpha, \beta \& \gamma$ are the roots of the equation $x^3+px+q=0$,

then the value of the determinant $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} =$

(A) p

(B) q

(C) p^2-2q

(D) none

Q6 If the system of equations

$$x+2y+2z=1$$
$$x-y+3z=3$$
$$x+11y-z=b$$

has solutions, then the value of b lies in the interval-

(A) $(-7,-4)$ (B) $(-4,0)$

(C) $(0,3)$ (D) $(3,6)$

Q7 The system of equations

$$kx+(k+1)y+(k-1)z=0$$
$$(k+1)x+ky+(k+2)z=0$$
$$(k-1)x+(k+2)y+kz=0$$

has a non-trivial solution for-

(A) exactly three real values of k.

(B) exactly two real values of k.

(C) exactly one real value of k.

(D) infinite number of values of k.

Q8 If A, B and C are such that $A+B+C=0$, then the value of $\begin{vmatrix} 1 & \cos C & \cos B \\ \cos C & 1 & \cos A \\ \cos B & \cos A & 1 \end{vmatrix}$ is

(A) 1

(B) $1-\cos^2 A-\cos^2 B-\cos^2 C-2\cos A\cos B\cos C$

(C) 0

(D) -1

Q9 The determinant $\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix}=0$ if:

(A) x, y, z are in A.P.

(B) x, y, z are in G.P.

(C) x, y, z are in H.P.

(D) xy, yz, zx are in A.P.

Q10 Area of triangle whose vertices $(a, a^2)\ (b, b^2)\ (c, c^2)$ is $\frac{1}{2}$, and area of another triangle whose vertices are $(p, p^2), (q, q^2)$ and (r, r^2) is 4, then the value of $\begin{vmatrix} (1+ap)^2 & (1+bp)^2 & (1+cp)^2 \\ (1+aq)^2 & (1+bq)^2 & (1+cq)^2 \\ (1+ar)^2 & (1+br)^2 & (1+cr)^2 \end{vmatrix}$,

(A) 2 (B) 4

(C) 8 (D) 16

Q11 If the system of equations

$$\left.\begin{aligned} x-\lambda y-z=0 \\ \lambda x-y-z=0 \\ x+y-z=0 \end{aligned}\right\}$$

has a unique solution, then the range of λ is $\mathrm{R}-\{\mathrm{a}, \mathrm{b}\}$. Then the value of $(\mathrm{a}^2+\mathrm{b}^2)$ is:

(A) 1 (B) 2

(C) 4 (D) 9

Q12 The value of $\begin{vmatrix} -1 & 2 & 1 \\ 3+2\sqrt{2} & 2+2\sqrt{2} & 1 \\ 3-2\sqrt{2} & 2-2\sqrt{2} & 1 \end{vmatrix}$ is equal to

(A) zero

(B) $-16\sqrt{2}$

(C) $-8\sqrt{2}$

(D) None of these

Q13 Let α, β be the roots of the equation $\mathrm{ax}^2+\mathrm{bx}+\mathrm{c}=0$. Let $\mathrm{S}_\mathrm{n}=\alpha^\mathrm{n}+\beta^\mathrm{n}$ for $\mathrm{n} \geq 1$ and

$$\Delta=\begin{vmatrix} 3 & 1+\mathrm{S}_1 & 1+\mathrm{S}_2 \\ 1+\mathrm{S}_1 & 1+\mathrm{S}_2 & 1+\mathrm{S}_3 \\ 1+\mathrm{S}_2 & 1+\mathrm{S}_3 & 1+\mathrm{S}_4 \end{vmatrix}$$

If $\Delta<0$, then the equation $\mathrm{ax}^2+\mathrm{bx}+\mathrm{c}=0$ has

(A) positive real roots

(B) negative real roots

(C) equal roots

(D) imaginary roots

Q14 Let A and B be two invertible matrices of order 3×3. If det $(\mathrm{ABA}^\mathrm{T})$ = 8 and det (AB^{-1}) = 8, then det $(\mathrm{BA}^{-1}\mathrm{B}^\mathrm{T})$ is equal to:

(A) $\frac{1}{4}$ (B) 1

(C) $\frac{1}{16}$ (D) 16

Q15 If $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix}=\frac{9}{8}\left(103x+81\right)$, then $\lambda, \frac{\lambda}{3}$ are the roots of the equation

(A) $4x^2+24x-27=0$ (B) $4x^2-24x+27=0$

(C) $4x^2+24x+27=0$ (D) $4x^2-24x-27=0$

Q16 If $\mathrm{a}_1, \mathrm{a}_2, \mathrm{a}_3, 5, 4, \mathrm{a}_6, \mathrm{a}_7, \mathrm{a}_8, \mathrm{a}_9$ are in $\mathbf{H}$. P. and $D=\begin{vmatrix} a_1 & a_2 & a_3 \\ 5 & 4 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$, then the value of $[D]$ is (where $[\cdot]$ represents the greatest integer function)

Q17 Let $D_1=\begin{vmatrix} a & b & a+b \\ c & d & c+d \\ a & b & a-b \end{vmatrix}$ and $D_2=\begin{vmatrix} a & c & a+c \\ b & d & b+d \\ a & c & a+b+c \end{vmatrix}$ then the value of $\left|\frac{D_1}{D_2}\right|$, where $b \neq 0$ and $ad \neq bc$, is.........

Q18 If $\Delta=\begin{vmatrix}1 & 3\cos\theta & 1\\ \sin\theta & 1 & 3\cos\theta\\ 1 & \sin\theta & 1\end{vmatrix}$, then the value of $\left(\Delta_{\max}\right)/2$ is.............

Q19 If $\begin{vmatrix}x^n & x^{n+2} & x^{n+4}\\ y^n & y^{n+2} & y^{n+4}\\ z^n & z^{n+2} & z^{n+4}\end{vmatrix}=\left(\frac{1}{y^2}-\frac{1}{x^2}\right)\left(\frac{1}{z^2}-\frac{1}{y^2}\right)\left(\frac{1}{x^2}-\frac{1}{z^2}\right)$ then $-\mathrm{n}$ is ....

Q20 If the system of linear equations

$$(\cos\theta)x+(\sin\theta)y+\cos\theta=0$$
$$(\sin\theta)x+(\cos\theta)y+\sin\theta=0$$
$$(\cos\theta)x+(\sin\theta)y-\cos\theta=0$$

is consistent, then find the number of possible value of θ is .(where $\theta\in[0,2\pi]$)

Q21 Assertion (A): If the matrix $A=\begin{bmatrix}1 & 3 & \lambda+2\\ 2 & 4 & 8\\ 3 & 5 & 10\end{bmatrix}$ is singular, then $\lambda=4$.

Reason (R): If A is a singular matrix, then $|A|=0$.

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is not the correct explanation of A

(C) A is true but R is false

(D) A is false and R is true

Q22 Let α,β be the roots of the equation $\mathrm{ax}^2+\mathrm{bx}+\mathrm{c}=0$.

Let $S_n=\alpha^n+\beta^n$ for $n\geq 1$ and

$$\Delta=\begin{vmatrix}3 & 1+S_1 & 1+S_2\\ 1+S_1 & 1+S_2 & 1+S_3\\ 1+S_2 & 1+S_3 & 1+S_4\end{vmatrix}$$

If $\mathrm{a},\mathrm{b},\mathrm{c}$ are rational and one of the roots of the equations is $1+\sqrt{2}$, then the value of Δ is

(A) 8 (B) 12

(C) 30 (D) 32

Q23 Assertion (A): If every element of a third order determinant of value D is multiplied by 5 , then the value of the new determinant is $125D$.

Reason (R): If k is a scalar and A is an $n\times n$ matrix, then $|kA|=k^n|A|$

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is not the correct explanation of A

(C) A is true but R is false

(D) A is false and R is true

Q24 The system of equation

$$x+3y+2z=6$$
$$x+ay+2z=7$$
$$x+3y+2z=b$$

has

(A) unique solution, if $\mathrm{a}=2$ and $\mathrm{b}\neq 6$.

(B) infinitely many solution, if $\mathrm{a}=4$ and $\mathrm{b}=6$.

(C) no solution, if $\mathrm{a}=5$ and $\mathrm{b}=7$.

(D) no solution, if $\mathrm{a}=3$ and $\mathrm{b}=5$.

Q25 Which of the following determinant(s) vanish(es)?

(A) $\begin{vmatrix}1 & bc & bc(b+c)\\ 1 & ca & ca(c+a)\\ 1 & ab & ab(a+b)\end{vmatrix}$

(B) $\begin{vmatrix}1 & ab & \frac{1}{a}+\frac{1}{b}\\ 1 & bc & \frac{1}{b}+\frac{1}{c}\\ 1 & ca & \frac{1}{c}+\frac{1}{a}\end{vmatrix}$

(C) $\begin{vmatrix}0 & a-b & a-c\\ b-a & 0 & b-c\\ c-a & c-b & 0\end{vmatrix}$

(D) $\begin{vmatrix}\log_x xyz & \log_x y & \log_x z\\ \log_y xyz & 1 & \log_y z\\ \log_z xyz & \log_z y & 1\end{vmatrix}$

Q26 A and B are square matrices such that $\det.(A)=1, BB^T=I,$ $\det.(B)>0,$ and $A(\operatorname{adj}.A+\operatorname{adj}.B)=B.$ $AB^{-1}=$

(A) $B^{-1}A$

(B) AB^{-1}

(C) A^TB^{-1}

(D) $B^T A^{-1}$

Q27 Let α, β be the roots of the equation $\mathrm{ax}^2+\mathrm{bx}+\mathrm{c}=0$. Let $\mathrm{S}_{\mathrm{n}}=\alpha^{\mathrm{n}}+\beta^{\mathrm{n}}$ for $\mathrm{n} \geq 1$ and

$$\Delta=\left|\begin{array}{ccc}3 & 1+\mathrm{S}_1 & 1+\mathrm{S}_2 \\ 1+\mathrm{S}_1 & 1+\mathrm{S}_2 & 1+\mathrm{S}_3 \\ 1+\mathrm{S}_2 & 1+\mathrm{S}_3 & 1+\mathrm{S}_4\end{array}\right|$$

If $\Delta>0$, then

(A) $\mathrm{f}(1)>0$

(B) $\mathrm{f}(1)<0$

(C) $\mathrm{f}(1)=0$

(D) cannot say anything about $\mathrm{f}(1)$

Q28 If

$$\left|\begin{array}{ccc}x & 3 & 6 \\ 3 & 6 & x \\ 6 & x & 3\end{array}\right|=\left|\begin{array}{ccc}2 & x & 7 \\ x & 7 & 2 \\ 7 & 2 & x\end{array}\right|=\left|\begin{array}{ccc}4 & 5 & x \\ 5 & x & 4 \\ x & 4 & 5\end{array}\right|=0,$$

then x is equal to

(A) 0

(B) -9

(C) 3

(D) none of these

Q29 If a, b, c are positive and are the p th, q th, and r th terms, respectively, of a G.P., then $\Delta=\left|\begin{array}{ccc}\log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1\end{array}\right|$ is

(A) 0

(B) $\log (a b c)$

(C) $-(p+q+r)$

(D) none of these

Q30 If $\left|\begin{array}{ccc}a & b-c & c+b \\ a+c & b & c-a \\ a-b & a+b & c\end{array}\right|=0$, then the line $a x+b y+c=0$ passes through the fixed point which is

(A) $(1,2)$

(B) $(1,1)$

(C) $(-2,1)$

(D) $(1,0)$

Answer Key

Q1	(A)
Q2	(A)
Q3	(C)
Q4	(C)
Q5	(D)
Q6	(A)
Q7	(C)
Q8	(C)
Q9	(B)
Q10	(D)
Q11	(B)
Q12	(B)
Q13	(D)
Q14	(C)
Q15	(B)
Q16	2
Q17	2
Q18	5
Q19	4
Q20	2
Q21	(A)
Q22	(D)
Q23	(A)
Q24	(B, C, D)
Q25	(A, B, C, D)
Q26	(A)
Q27	(D)
Q28	(B)
Q29	(A)
Q30	(B)